

ETHERREAL VS. TCPDUMP: A COMPARATIVE STUDY ON PACKET SNIFFING TOOLS FOR EDUCATIONAL PURPOSE*

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ABSTRACT

There are many free packet-sniffing tools available for download. Ethereal and tcpdump are two of the most popular tools among network administrators. This work compares and contrasts the usefulness and appropriateness of these tools for pedagogical purposes. While ethereal is user-friendlier than tcpdump, tcpdump is less intrusive and hence, can be used in a campus-wide network safer, since it does not readily reveal any data transmitted in a packet. Ethereal can be used in a closed networking lab environment to analyze and study many more protocols. Many class assignments can be designed using these two packet sniffers. Particularly, assignments can be developed to analyze tcpdump's output in real-time for intrusion detection or the understanding of a protocol.

1. INTRODUCTION

Packet sniffers are used by network administrators for many reasons including: converting binary data in packets to human readable format, troubleshooting problems on a network, network intrusion detection, logging network traffic for forensics and evidence, and discovery of a faulty network card [4]. Packet sniffers can also be used for educational purposes in a computer networking course. In particular, a packet sniffer can be used to observe the sequence of packets exchanged between a sender and a receiver and hence, an in-depth understanding of the underlying protocol used in the communication can be gained from such observation. Some packet sniffers allow only

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viewing of the header fields of a packet, while others allow even viewing the contents of a packet. A packet sniffer can be used for positive purposes as well for malicious purposes. Many full-featured packet sniffers are available for download free of charge [1]. As a result, hackers have free access to them. A hacker or an intruder can use a packet sniffer directly or indirectly in many different ways, including the capturing of cleartext usernames and passwords, compromising proprietary information, mapping out a network, and breaking into a network.

A packet sniffer is rarely the only tool used for an attack. The reason is that a sniffer only works in what is known as a common collision domain on an Ethernet, for example. A common collision domain is a network segment that is connected through a hub, not switched or bridged. With the use of a switch, all hosts on the segment can see any traffic on their own segment but cannot see the traffic in other segments in the network. While switches limit the scope of a packet sniffer in a network, hubs expand it. Figure 1 shows a switched network that connects segments X, Y, and Z. A packet sniffer running on a host on segment X will not see any traffic from segment Y or Z except for broadcast traffic or traffic aimed at any host on segment X. If all the segments were connected to each other through a hub, rather than a switch, a packet sniffer on segment X would see traffic in segments Y and Z as well.

Computer security can be compromised in many ways, for example, someone walks in with a laptop and plugs into a network. For comments sake, concerning wireless LAN (Local Area Network) technology, someone sitting in the parking lot with a laptop and a wireless NIC (Network Interface Card) can connect to a network in the same way as if he/she walked through the front door and plugged directly into a hub and watch traffic. To guard against this sort of security breach, it is important to train students, our future networking professionals, with the tools and techniques available today which might be used by a hacker illegally.

The main objective of this paper is to present how packet sniffers can be used as educational resources to students in a computer networking class. Among freely available packet sniffers, particularly, tcpdump and ethereal have gained popularity as network

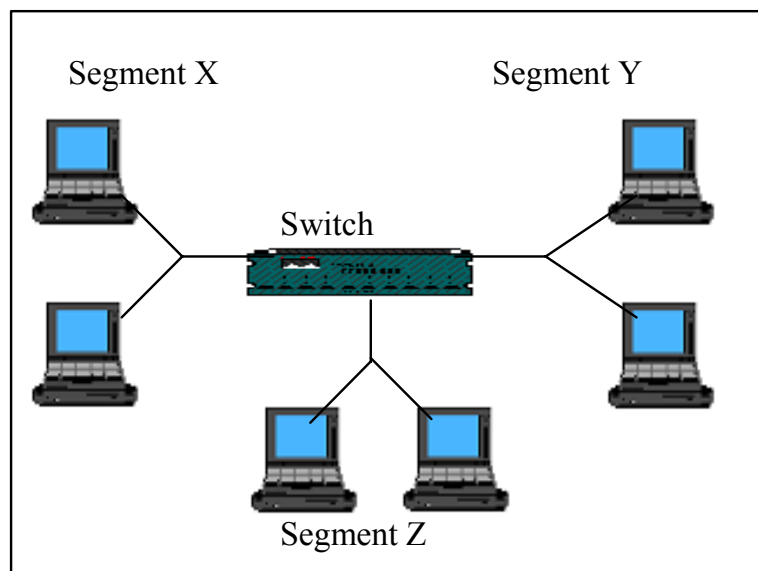


Figure 1. A Switched Network

analyzing tools to network administrators and security professionals. It is important to know, compare, and contrast the usefulness, strengths, and weaknesses of these tools in analyzing packets for educational purposes. The rest of the paper is organized as follows. In section 2, a brief introduction to TCP/IP (Transmission Control Protocol/Internet Protocol) is given. Section 3 presents how to use tcpdump and ethereal for capturing packets for various experimental purposes. In section 4, both tcpdump and ethereal are compared for their usefulness. Section 5 briefly outlines some example experiments using tcpdump and ethereal. Section 6 concludes the paper with some cautionary remarks on the use of the tools in a public network.

2. A BRIEF INTRODUCTION TO TCP/IP

TCP/IP has become the standard protocol for communication between computers on the Internet. IP (Internet Protocol) provides functionality at the network layer (addressing and routing) while TCP (Transmission Control Protocol) provides (virtual) end-to-end connectivity between two applications running two end-hosts. The TCP/IP family includes a host of other useful protocols such as ICMP (Internet Control Message Protocol), IGMP (Internet Group Management Protocol), and UDP (User Datagram Protocol). An overwhelming majority of today's networks use TCP/IP. Almost every other application today incorporates some kind of network functionality; hence it has become necessary for every programmer to have at least a working knowledge of TCP/IP. Communication between computers using TCP/IP takes place through the exchange of packets. A packet is a PDU (Protocol Data Unit) at the IP layer. The PDU at the TCP layer is called a segment while a PDU at the data-link layer (such as Ethernet) is called a frame. However, the term packet is generically used to describe the data unit that is exchanged between TCP/IP layers as well as between two computers. A packet captured on the Internet can contain a transport layer header, a network layer header, and a data-link layer header and a payload from a message. A packet analyzing tool can provide detail information on each header part as well as on the payload part.

3. NETWORK PROTOCOL ANALYZERS

Network protocol analyzers, such as tcpdump and ethereal, work by setting the network interface into a mode, called promiscuous mode, in which the network interface card captures all traffic that pass that interface.

3.1 TCPDUMP

Tcpdump was developed in the early 1990s [5] at the Lawrence Berkeley National Laboratory. Tcpdump is a UNIX tool (WinDump for windows) used to gather data from a network, decipher the bits and pieces, and display them in a semi-coherent fashion. Although, output from commercial tools might differ slightly or be more fashionable than tcpdump, tcpdump is the basis that most packet sniffing software are comprised of and hence, can aid in learning other tools, such as ethereal. Figure 2 shows the first two lines of a tcpdump output for an ftp (file transfer protocol) session from one interface to another on a network segment.

```
[root@localhost ialabs]# tcpdump -n host 10.0.1.12
tcpdump: listening on eth0

15:14:54.281059 10.0.1.11.33129 > 10.0.1.12.ftp: S
2860245706:2860245706(0) win 5840 <mss 1460,sackOK,timestamp
1887180,nop,wscale 0> (DF)

15:14:54.281416 10.0.1.12.ftp > 10.0.1.11.33129: S 3866158800:3866158800(0) ack 2860245707 win 5792
<mss 1460,sackOK,timestamp 404001 188718,nop,wscale
```

Figure 2. An ftp session captured by tcpdump.

Tcpdump displays one line for each transmitted or received frame. In each line, tcpdump displays a timestamp and information that is derived from the protocol headers contained in a frame. As shown in figure 2 in boldface font, the timestamp 15:14:54.281059 corresponds to 3:14PM and 54.281059 seconds. If the frame is an IP Datagram with a UDP or TCP payload, then tcpdump displays information on the source and destination of the frame. The entry 10.0.1.11.33129 in the first line of output indicates that the sender's IP address is 10.0.1.11 and its application port number is 33129 and the destination address is 10.0.1.12 and its application port is the default ftp port. The symbol "S" in the packet represents "SYN," or the synchronization flag in the TCP header of the packet, meaning it is a synchronization packet. The entry "2860245706:2860245706(0)" in the tcpdump output means the sending TCP stack is setting 2860245706 as the initial synchronization number (ISN), and "0" (no) data is being passed in this packet. Although the numbers before and after the colon (:) are the same for this packet, in later packets, they will be different and will have explanatory value. The entry "win 5840 < mss 1460" means the TCP available window size is 5840 bytes and maximum segment size of 1460 bytes. The remainder of the line includes nop (no option) padding and selective acknowledgement described in Request for Comments 2018 (RFC 2018) [6].

As can be seen, the process of dissecting a tcpdump "dump" can become rather tedious very quickly. By simply just issuing the tcpdump command, all traffic on that interface will be "dumped" to standard output. Instead of capturing all traffic and then searching through the output for the data of interest, tcpdump allows limiting of traffic to be captured by specifying a filter expression. With a filter expression, only the traffic that matches the filter expression is captured and displayed.

3.2. ETHEREAL

Gerald Combes developed ethereal in 1997. Not long after, Gilbert Ramirez of Cisco Systems saw its potential and contributed a low-level dissector to it [2]. In October 1998, Guy Harris of Network Appliance started applying patches and contributing dissectors to ethereal as well. In late 1998, Professor Richard Sharpe saw its potential on packet analysis courses and also contributed dissectors and patches to ethereal [2]. The list of people who have contributed to ethereal has become very long since then.

Ethereal is free, available for both Windows and Unix, and is particularly user-friendly because it has a graphical user interface to assist in navigating the capture and analysis of packets. Ethereal can read tcpdump binary output and also use tcpdump

filters to selectively capture or display records. Ethereal offers more protocol decoding options, more than four hundred in all [3], than any free sniffer available and ranks well among commercial tools. Ethereal was built on the fundamental properties of tcpdump (filtering, etc.), but ethereal makes it easier to make sense of a stream of ongoing network communications. Ethereal not only makes network troubleshooting work far easier, but it also aids greatly in network forensics, the art of finding and examining an attack, by giving the analyst a better view.

Figures 3 and 4 below show the output of ftp and secure shell sessions analyzed by ethereal. As can be seen, the output is more organized and detailed than the output of tcpdump. A capability of ethereal is its ability to display the login username and password of ftp and telnet sessions as shown in figure 3, implying that ftp and telnet sessions should

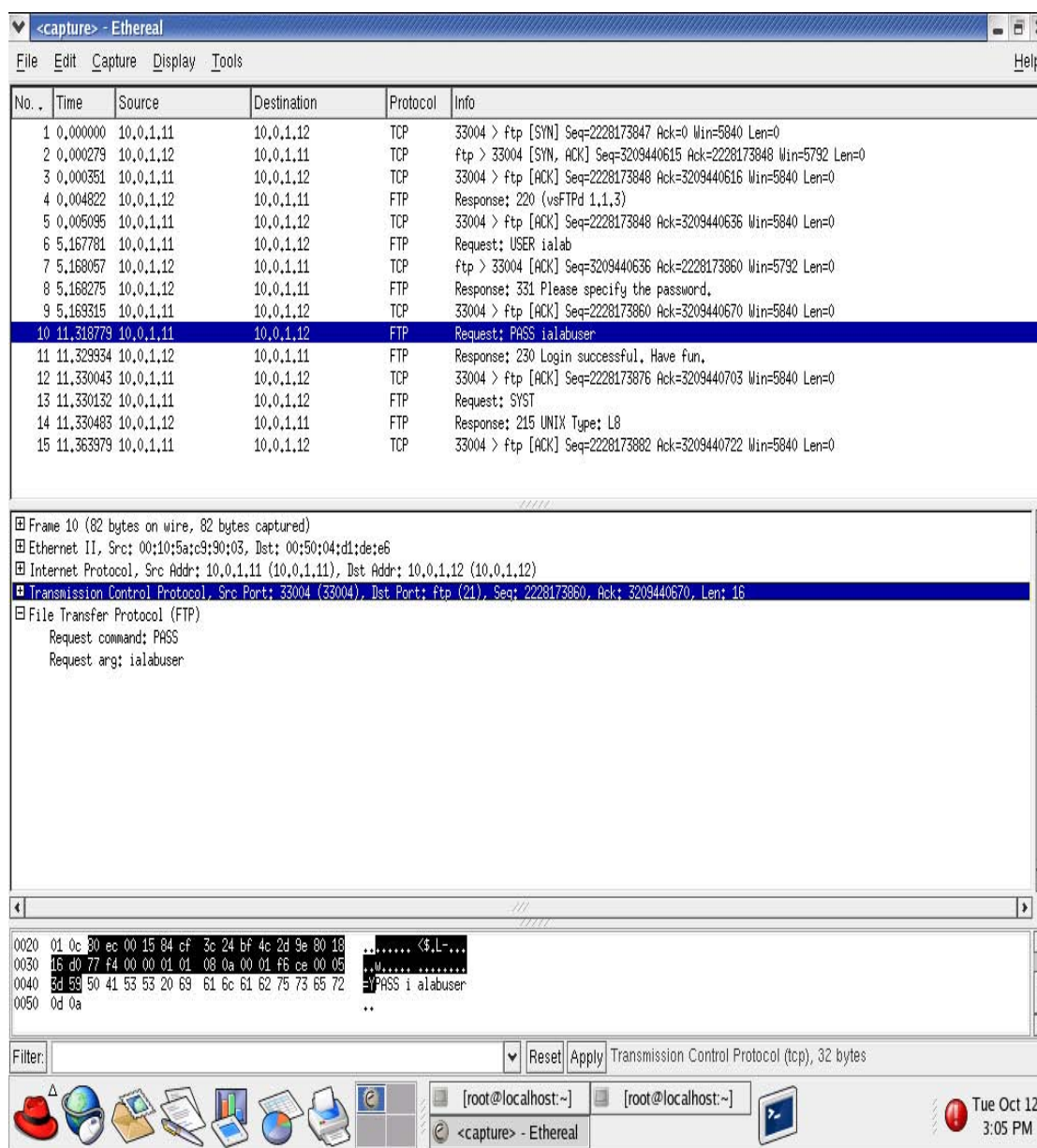


Figure 3. An ftp login session captured by ethereal.

be avoided when logging in as a root user. Figure 4 shows the inability of ethereal to decrypt the login process in secure shell.

Once the traffic capture is started, the ethereal window displays the traffic in three different views. The first view shows a summary of the captured packets, one line for each packet. One of these packets can be highlighted, by clicking on the corresponding line. The second view shows the protocol header, which details the highlighted packet. Packet headers are displayed in a tree-like fashion and can be minimized or maximized. The third view shows the ASCII and hexadecimal representation of the packet headers and the data. Like tcpdump, ethereal also has a rich filtering language to capture only the desired packets that match a specified filter.

4. COMPARISON

Ethereal has a very good graphical user interface, can provide information on packet basis or protocol basis, and can display packet data in meaningful format. Hence it is a very popular tool among network administrators. The user interface of ethereal is well designed so that even a novice can understand which packets are getting captured and what information they contain. The good interface makes the learning process enjoyable. On the other hand, tcpdump does not have any graphical user interface and primarily provides information on packet headers only, but it does not display data in a packet. Thus, we observe that tcpdump is less intrusive than ethereal. For educational purposes, allowing students to run ethereal in a campus-wide network is not a good idea because

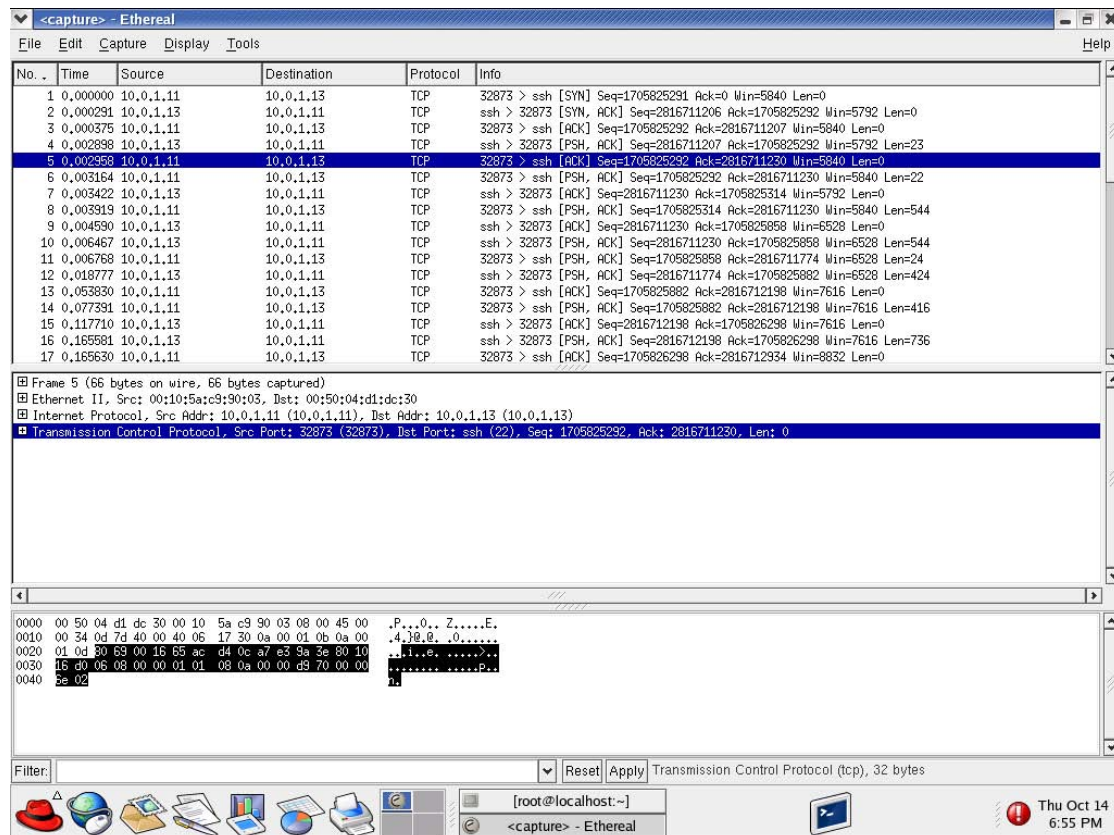


Figure 4. A secure shell session captured by ethereal.

a student can easily view users' passwords on the network for some protocols such as ftp, telnet, and pop3. Ethereal should be run in a closed lab environment on an experimental network. In this regard, tcpdump is better than ethereal and can be run less restrictively on a campus-wide network. In addition, it is important for the beginners to become familiar with some bare-bone packet-analyzing tool like tcpdump for better understanding of a protocol without compromising security. Students can be introduced with tcpdump first and then with ethereal next in a closed lab network.

While tcpdump is an extremely good tool, it focuses mainly on TCP/IP protocol and does its job well. Ethereal is much more versatile and can understand a variety of protocols. Neither ethereal or tcpdump alone act as an intrusion detection system. Neither will warn when an intruder does strange things to a network. However, when strange things do happen, they might help the analyst figure out what is really going on. Also, neither can manipulate data on a network, they will only "measure" the data on it. Neither are able to send packets on a network or do other active things, except for name resolutions, but even that can be disabled. The usefulness of tcpdump, however, can be extended by writing a script (or a program) that would take tcpdump's output as input (through piping) for real-time analysis for intrusion detection or similar purposes. Based on this scheme, many class assignments can be designed using tcpdump.

5. LAB EXPERIMENTS WITH ETHEREAL AND TCPDUMP

A lab on ethereal or tcpdump can be developed as a sequence of three components such as pre-lab, in-lab, and post-lab components. For the pre-lab component, students may be asked to write expressions to filter captured traffic data. Here are some examples on how to write expressions to filter packets using ethereal:

- 1) Write an expression for an ethereal display filter to show all IP datagrams that have 10.0.1.50 in its destination IP address field and the frame size greater than 400 bytes.

`ip.dst == 10.0.1.50 and frame.pkt_len > 400`

- 2.) Write an expression for an ethereal display filter to show the packets containing an ICMP message with the source or destination IP address as 10.0.1.12 and the frame number of which is between 15 and 30.

`icmp and ip.addr == 10.0.1.12 and frame.number > 15 and frame.number < 30`

Students can verify their pre-lab work with the actual in-lab work later. A similar pre-lab could be assigned to students to introduce tcpdump. Here are some pre-lab problems on tcpdump:

- 1) Write the tcpdump command that will capture any packets exchanged between two hosts with the IP addresses 10.0.1.11 and 10.0.1.12.

`tcpdump -n host 10.0.1.11 and 10.0.1.12`

- 2) Write a tcpdump filter expression that will capture packets containing the source or destination IP address as 10.0.01.12.

`tcpdump -n tcp and host 10.0.1.12`

More complex experiments, using tcpdump and ethereal, can be designed as well. For example, using both tcpdump and ethereal, one can capture statistics associated with a

data transfer from one machine to another and then write a piece of code to parse and analyze the captured data to study the characteristics (throughput, path bandwidth, average queue delay, etc.) of the path between the two hosts on the Internet.

6. CONCLUSION

A packet sniffer can easily be seen as a double-edged sword. It can be used as an extraordinary tool for network analysis or as a hacking tool which poses a threat. Tcpdump and ethereal are excellent tools to supplement the theory discussed in an introductory computer networking class for understanding of network protocols. Both encourage an interactive experimentation to analyzing packets on a network. Ethereal's simple graphical interface provides an ease into the world of packet dissection while tcpdump displays the raw packet headers at the command prompt. Ethereal and tcpdump both can be used to develop many class assignments. However, ethereal should be used in a closed networking lab environment rather than in a campus-wide network since it can expose passwords or any other private information communicated by the campus users.

One final word to say about packet analyzers, the use of packet sniffers should be exercised with care, since using these tools in a publicly accessible network without authorization is not permitted. Students should be made aware about this fact.

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